- Whenever you visit a website, the domain name appears in the address bar of the web browser. Some domain names are preceded by "www" (which is not part of the domain name), while others omit the "www" prefix.

- All domain names have a domain suffix, such as .com, .net, or .org. The domain suffix helps identify the type of website the domain name represents. There are only a limited number of such domains. For example:

    - gov - Government agencies
    - edu - Educational institutions
    - org - Organizations (nonprofit)
    - mil - Military
    - com - commercial business
    - net - Network organizations
    - ca - Canada
    - th - Thailand

Domain names are relatively cheap to register, though they must be renewed every year or every few years. Anyone can register a domain name, so you can purchase a unique domain name for your blog or website.

When you access a website, the domain name is actually translated to an IP address, which defines the server where the website is located. This translation is performed dynamically by a service called DNS.

MAC Address

- MAC stands for "Media Access Control".
- A MAC address is a hardware identification number that uniquely identifies each device on a network.

IP address

- IP stands for "Internet Protocol." An IP address, is a unique address that identifies a device on the Internet or a local network.

A valid IP address must be in the form of xxx.xxx.xxx.xxx, where xxx is a number from 0-255. IPv6 is the sixth revision to the Internet Protocol and the successor to IPv4. It functions similarly to IPv4 in that it provides the unique, numerical IP addresses necessary for Internet-enabled devices to communicate. it utilizes 128-bit addresses. IPv4 uses 32 bits for its Internet addresses.

Workshop-5

Write down the IP address of the PCs used in your computer laboratory.

14.8. Wireless/Mobile Communication

Wireless communications is a type of data communication that is performed and delivered wirelessly. This is a broad term that incorporates all procedures and forms of connecting and communicating between two or more devices using a wireless signal through wireless communication technologies and devices.